TELCO CUSTOMER CHURN

Exploratory Data Analysis Report

An analysis of customer attributes, service usage, and billing behaviour to identify the key drivers of churn

Dataset: Telco Customer Churn

Records: 7,043 | Attributes: 21

1. Executive Summary

This report presents an exploratory data analysis (EDA) of the Telco Customer Churn dataset, which contains 7,043 customer records across 21 attributes covering demographics, subscribed services, contract terms, billing methods, and churn status. The objective of the analysis is to understand the characteristics of customers who discontinue their service and to surface the factors most strongly associated with churn, in order to inform retention strategy.

Overall, 26.54% of customers in the dataset have churned. The analysis shows that churn is concentrated among customers on month-to-month contracts, customers with short tenure, customers who pay by electronic check, and customers who subscribe to Fiber optic internet without add-on protection services such as Online Security or Tech Support. A chi-square test of independence confirms that the relationship between contract type and churn is statistically significant.

Key Findings

- 26.54% of customers have churned; 73.46% remain active.
- Month-to-month contract customers churn at a much higher rate than customers on 1-year or 2-year contracts.
- Customers with low tenure (1–2 months) churn far more often than long-tenured customers; retention improves steadily with tenure.
- Fiber optic subscribers without Online Security or Tech Support show the highest churn among service segments.
- Electronic check is the payment method most associated with churn.
- Senior citizens churn at a proportionally higher rate than non-senior customers.
- Tenure and Total Charges are strongly positively correlated ($r = 0.83$), while tenure is negatively correlated with churn ($r = -0.35$).
- A chi-square test between Churn and Contract type returned $p \approx 5.86 \times 10^{-258}$, indicating an extremely significant association.

2. Dataset Overview

The dataset consists of 7,043 unique customer records with no duplicate rows and no duplicate customer IDs. It contains 21 columns spanning demographic attributes (gender, senior citizen status, partner, dependents), account information (tenure, contract type, payment method, paperless billing), subscribed services (phone, internet, security, backup, device protection, tech support, streaming TV and movies), billing figures (monthly and total charges), and the target variable, Churn.

Metric	Value

Total records	7,043
Total attributes	21
Duplicate rows	0
Duplicate customer IDs	0
Missing values	0 (after cleaning)
Churned customers	26.54%
Retained customers	73.46%

Data Cleaning Steps

- Blank string entries in TotalCharges were replaced with 0 and the column was converted from object to float type.
- The numeric SeniorCitizen flag (0/1) was recoded to categorical labels ("No"/"Yes") for consistency with other categorical fields and clearer visualisation.
- The dataset was checked for null values and duplicate records; none were found, confirming the data was already in a clean, analysis-ready state.

Descriptive Statistics - Numeric Fields

Statistic	Tenure (months)	Monthly Charges ($)	Senior Citizen
Mean	32.37	64.76	0.16
Std. Dev.	24.56	30.09	0.37
Minimum	0	18.25	0
25th percentile	9	35.50	0
Median	29	70.35	0
75th percentile	55	89.85	0
Maximum	72	118.75	1

3. Univariate Analysis

3.1 Customer Partnership Status

The customer base is fairly evenly split by partnership status, with a slightly larger share of customers reporting no partner (roughly 52%) compared to those with a partner (roughly 48%).

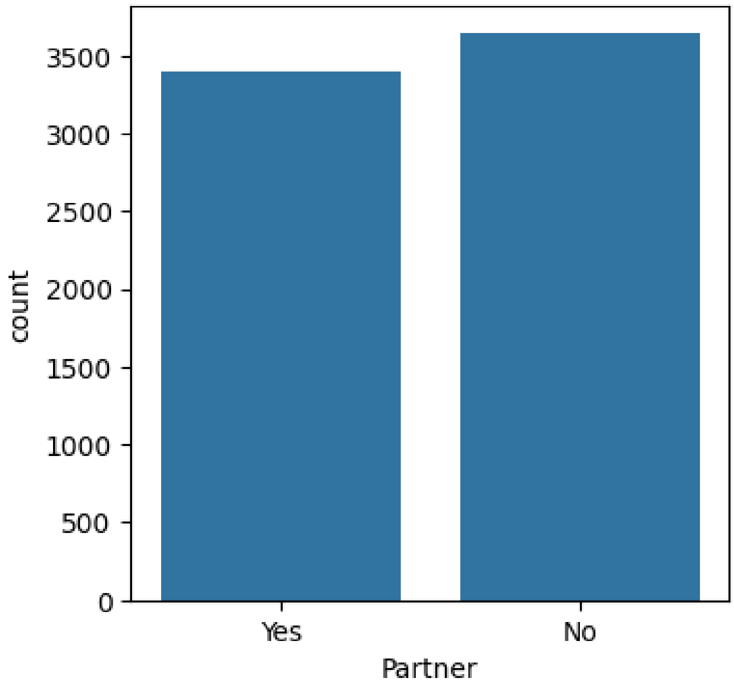

Figure 1. Count of customers by Partner status.

3.2 Overall Churn Distribution

Of the 7,043 customers in the dataset, 5,174 have remained active while 1,869 have churned. This establishes the baseline churn rate against which all subsequent segment comparisons are made.

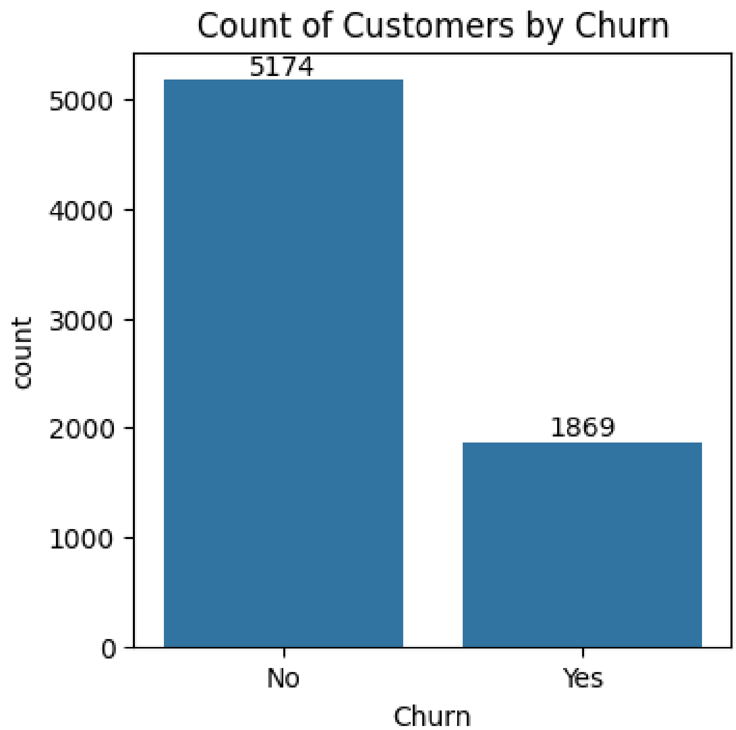

Figure 2. Count of customers by churn status.

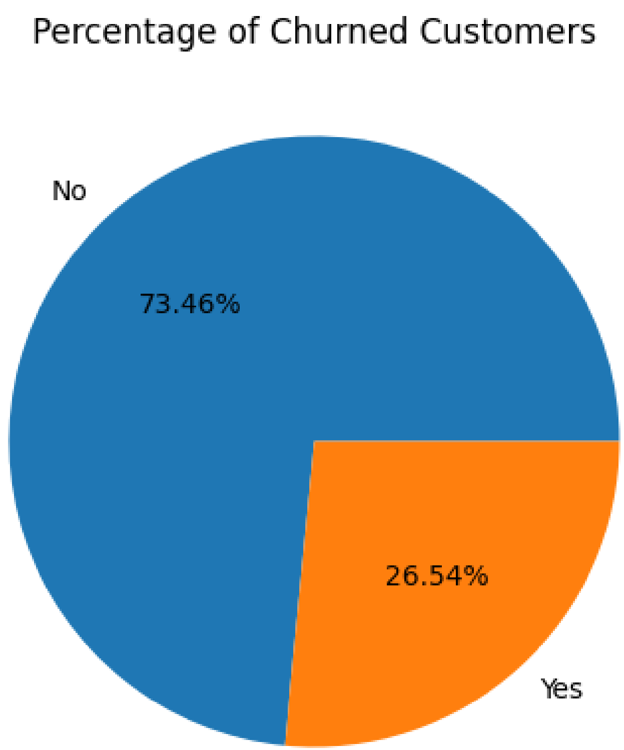

Figure 3. Churn distribution shown as a percentage. 26.54% of customers have churned.

4. Churn by Customer Segment

4.1 Gender

Churn is distributed almost identically between male and female customers, indicating that gender is not a meaningful driver of churn in this dataset.

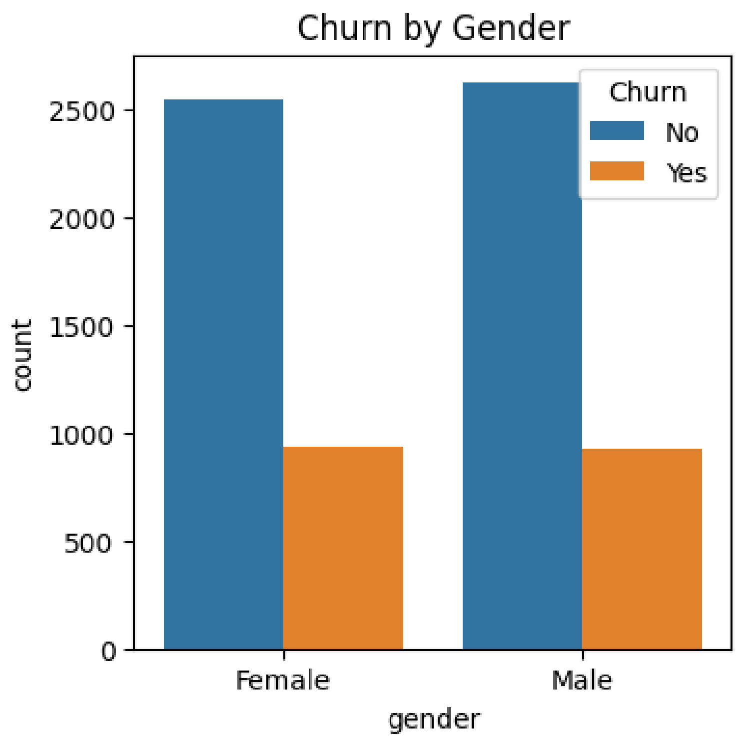

Figure 4. Churn counts split by gender.

4.2 Senior Citizen Status

Senior citizens make up a smaller share of the customer base but churn at a noticeably higher proportional rate than non-senior customers, suggesting age-related factors (e.g. service complexity or pricing sensitivity) may influence retention for this group.

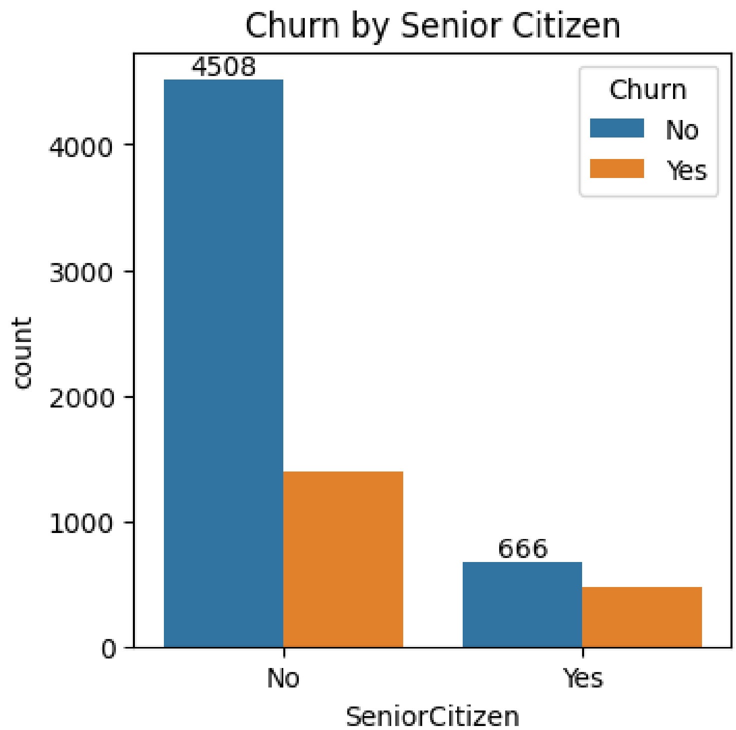

Figure 5. Churn counts split by senior citizen status.

4.3 Tenure

Churn is heavily concentrated among customers in their first 1–2 months of service. As tenure increases, the volume of churned customers drops sharply, while the count of retained customers grows toward the higher tenure bins. This pattern suggests that the first few months are the highest-risk window for customer attrition, and that customers who are retained past the early period are considerably more likely to remain long-term.

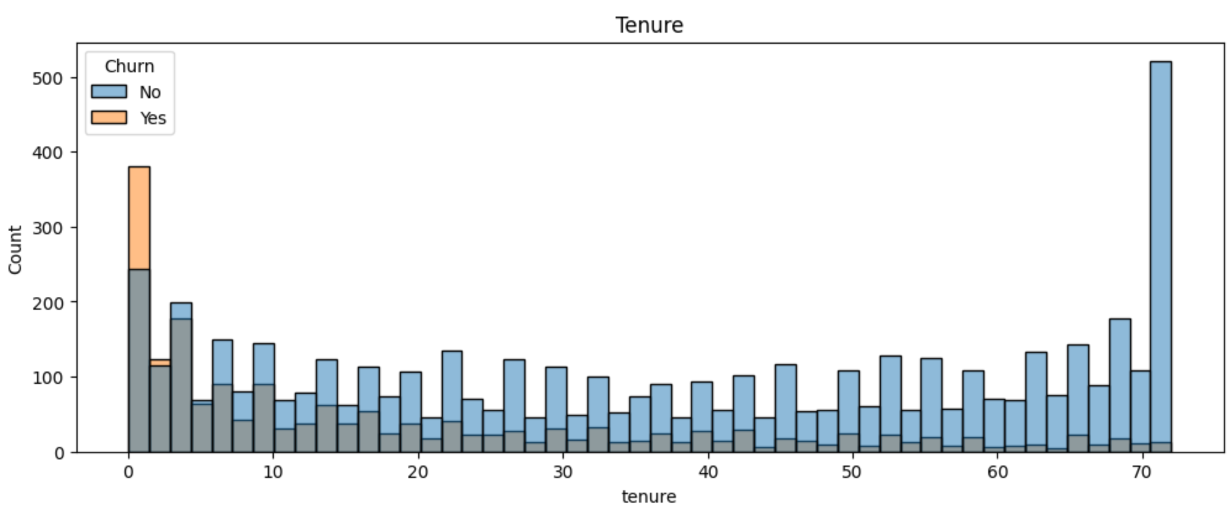

Figure 6. Distribution of customer tenure (in months), split by churn status.

4.4 Contract Type

Contract type shows one of the strongest relationships with churn in the dataset. Customers on month-to-month contracts churn at a substantially higher rate than those on one-year contracts, and two-year contract holders churn the least of all three groups. This indicates that longer contractual commitment is strongly associated with customer retention.

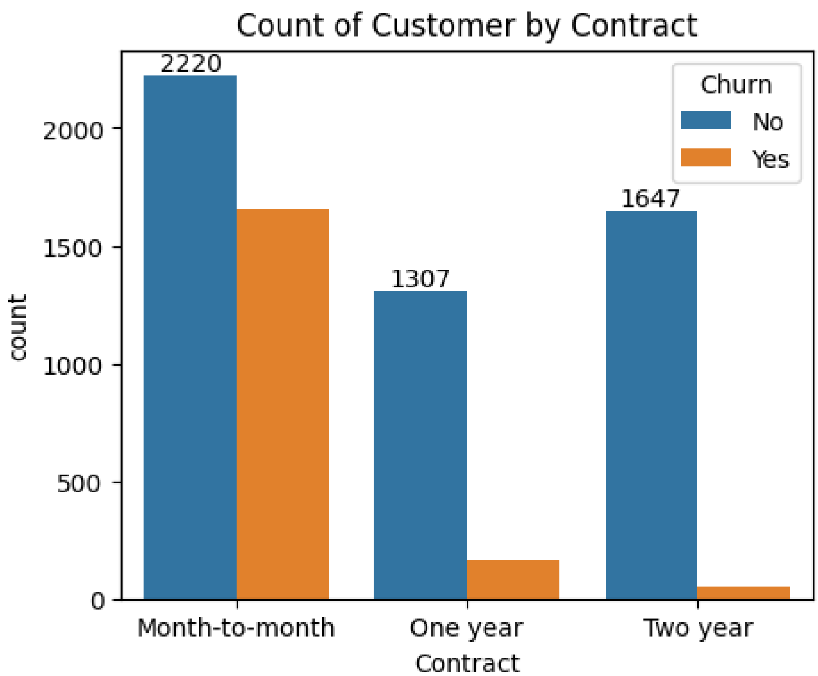

Figure 7. Count of customers by contract type, split by churn status.

5. Service Subscriptions and Churn

To understand how subscribed services relate to churn, nine service-related fields were examined together: Phone Service, Multiple Lines, Internet Service, Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, and Streaming Movies. Customers with Fiber optic internet, no Online Security, and no Tech Support show the highest churn rates. Basic services such as Phone Service alone are associated with much lower churn. In general, churn is more common among internet service subscribers, particularly those without value-added protection services.

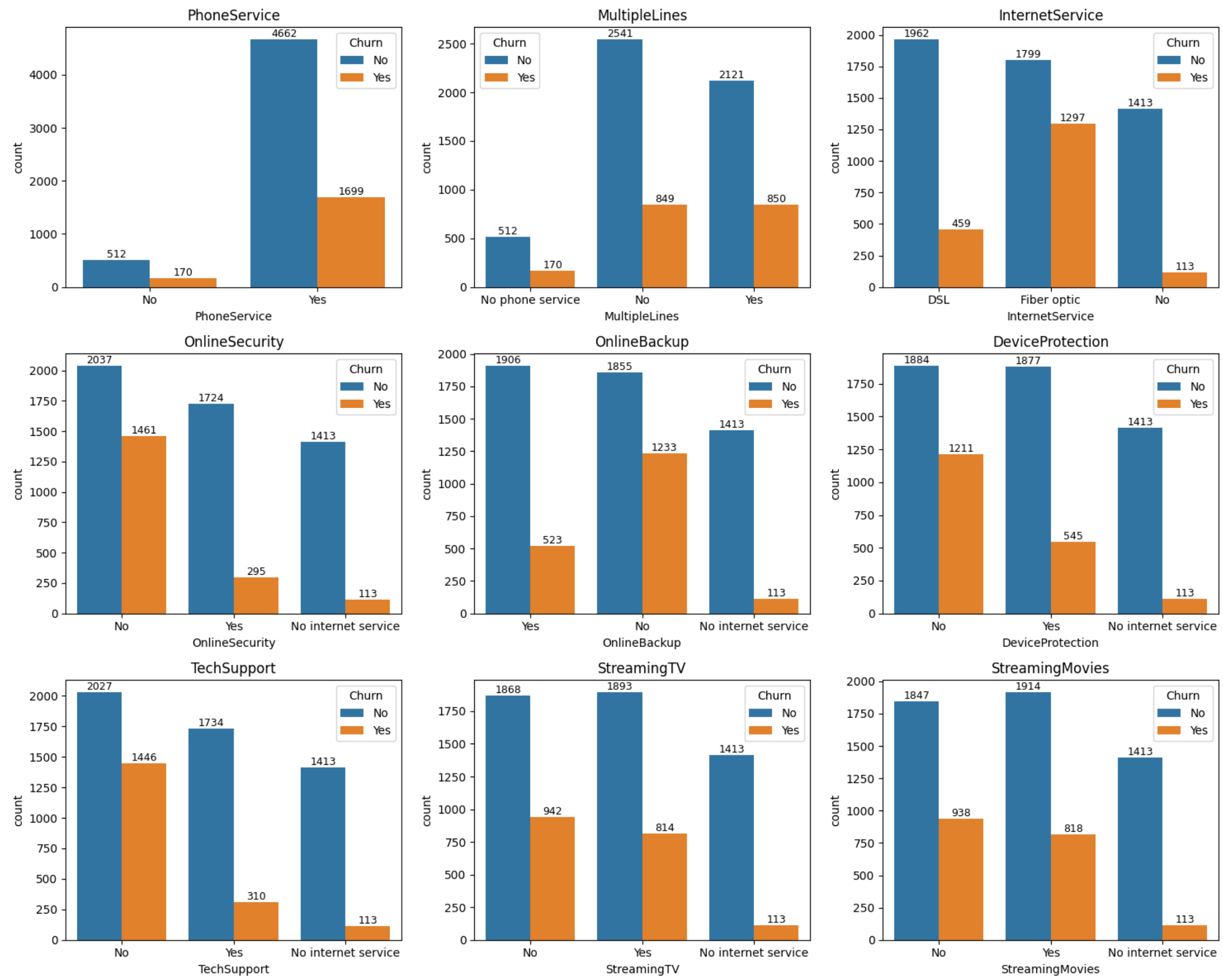

Figure 8. Churn breakdown across nine service subscription categories.

5.1 Payment Method

Payment method is another strong differentiator: customers who pay via Electronic Check churn considerably more often than those using automatic bank transfer, automatic credit card payment, or mailed cheque. This may reflect a segment of customers who are more price-sensitive or less engaged with the provider's automated billing relationship.

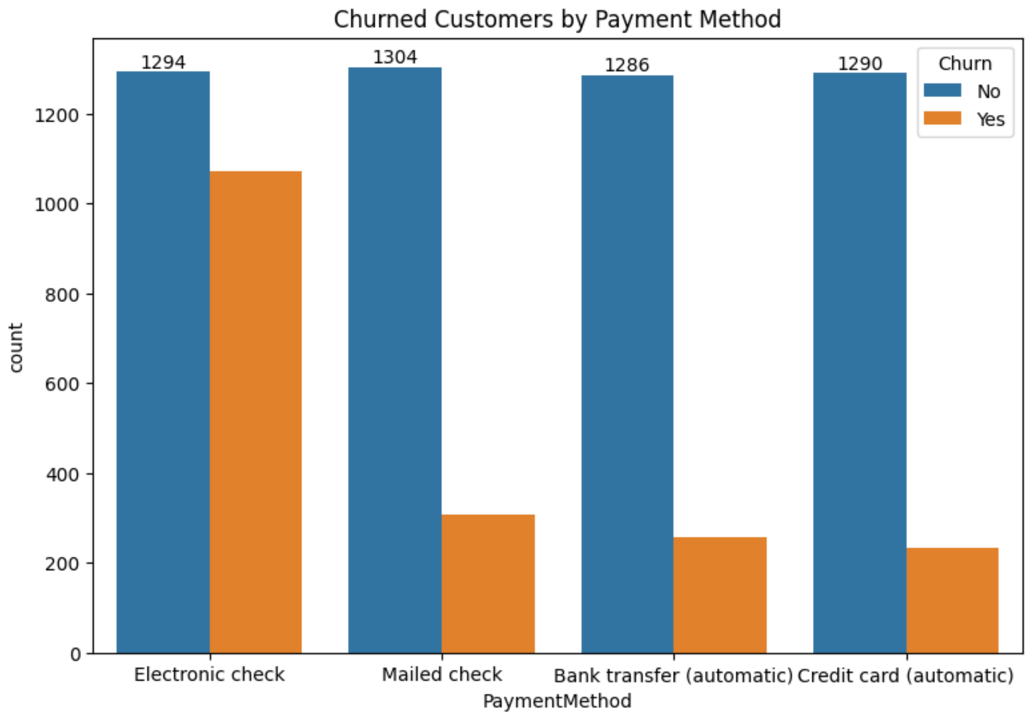

Figure 9. Count of churned and retained customers by payment method.

6. Billing and Financial Analysis

6.1 Average Monthly Charges vs. Tenure

Average monthly charges rise sharply over the first few months of tenure and then plateau at a relatively high, stable level for the remainder of the customer lifecycle. This suggests that customers who stay past the initial period tend to settle into a consistent, higher-value billing pattern.

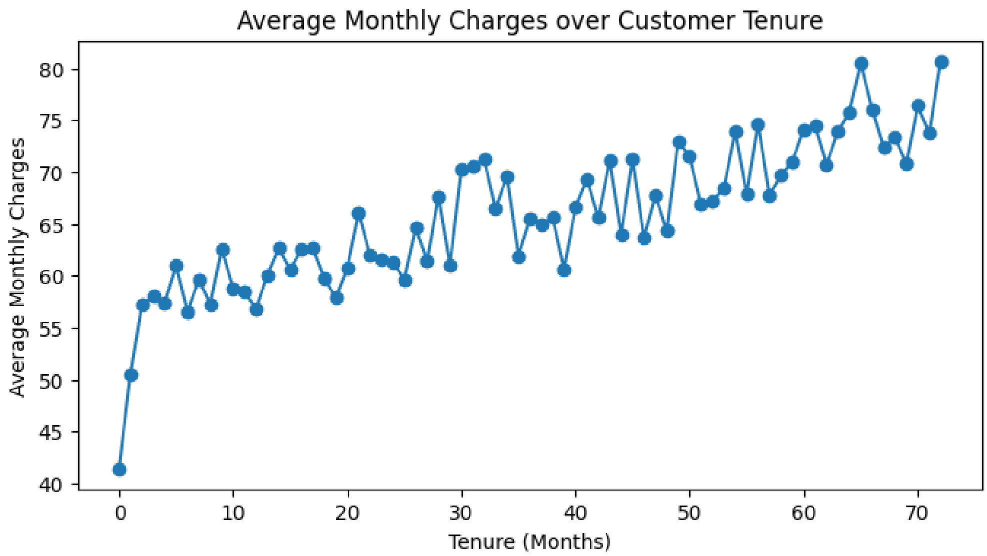

Figure 10. Average monthly charges plotted against customer tenure (months).

6.2 Monthly Charges vs. Total Charges

As expected, Total Charges increases with Monthly Charges and, more importantly, with tenure - since Total Charges accumulates over the customer relationship. Churned customers (shown in a distinct colour) are visibly concentrated in the lower-left region of the plot, corresponding to lower total charges and, by extension, shorter tenure, reinforcing the tenure-related findings above.

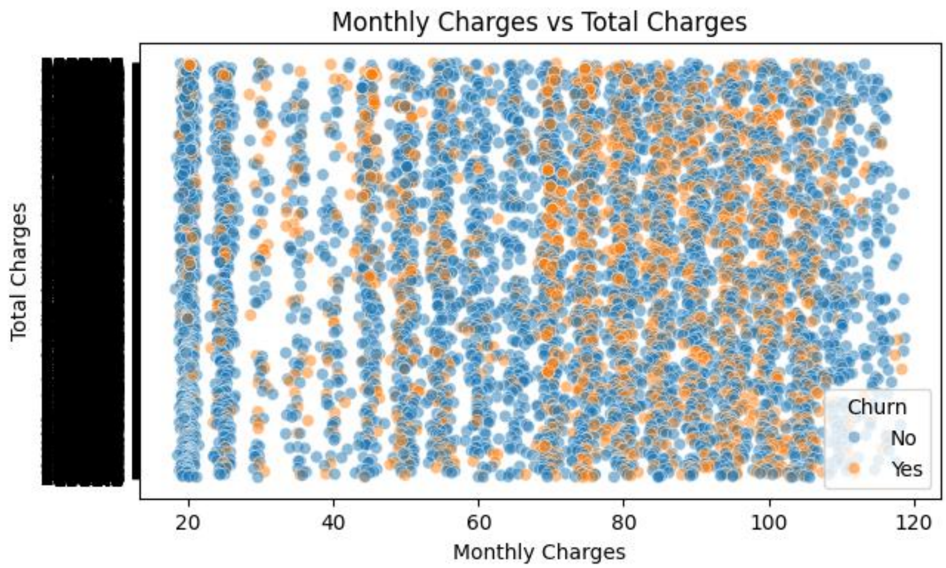

Figure 11. Scatter plot of Monthly Charges vs. Total Charges, coloured by churn status.

6.3 Correlation Analysis

A correlation heatmap was generated across the numeric fields (Senior Citizen, Tenure, Monthly Charges, Total Charges, and a binary-encoded Churn indicator) to quantify these relationships.

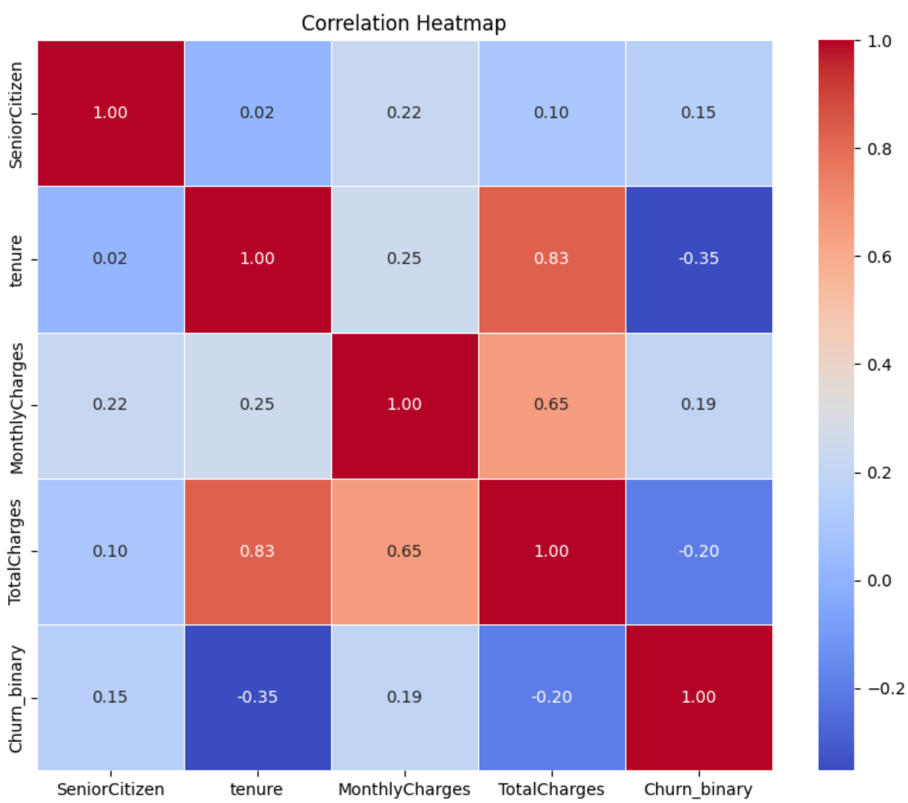

Figure 12. Correlation heatmap of numeric features.

Relationship	Correlation (r)	Interpretation
Tenure ↔ Total Charges	0.83	Strong positive - longer tenure drives higher cumulative billing
Monthly Charges ↔ Total Charges	0.65	Moderate positive
Tenure ↔ Churn	-0.35	Moderate negative - longer tenure reduces churn likelihood
Total Charges ↔ Churn	-0.20	Weak negative
Monthly Charges ↔ Churn	0.19	Weak positive
Senior Citizen ↔ Churn	0.15	Weak positive

7. Statistical Hypothesis Testing

To formally test whether contract type and churn are related, or whether any apparent relationship could be due to chance, a chi-square (χ^2) test of independence was performed on the contingency table of Churn versus Contract type.

Test	Variables	Result
Chi-square test of independence	Churn × Contract	$p \approx 5.86 \times 10^{-258}$

At any conventional significance threshold (e.g. $\alpha = 0.05$), this p-value is overwhelmingly smaller, leading to rejection of the null hypothesis of independence. This confirms, with very high statistical confidence, that contract type and churn are not independent - customers' contractual commitment level is a genuine and significant driver of churn behaviour, consistent with the visual pattern observed in Figure 7.

8. Conclusions and Recommendations

The exploratory analysis identifies a consistent and coherent churn profile: customers who are new (low tenure), on flexible month-to-month contracts, paying by electronic check, and using Fiber optic internet without add-on security or support services are the most likely to churn. These factors are mutually reinforcing - a new customer with no long-term contract and no added protections has few points of friction preventing them from leaving.

Recommendations

- Prioritise retention outreach and onboarding support during the first 1–2 months of tenure, the period of highest churn risk.
- Offer incentives (discounts, service credits) for customers to move from month-to-month to one- or two-year contracts.
- Investigate the Electronic Check payment experience and encourage migration to automatic payment methods, which are associated with lower churn.
- Bundle Online Security and Tech Support with Fiber optic plans, or promote their adoption among existing Fiber optic customers, to address the highest-churn service segment.
- Consider targeted retention programmes for senior citizens, who show a disproportionately higher churn rate.

These findings provide a strong foundation for a predictive churn model; the segments and relationships identified here (contract type, tenure, internet/service mix, and payment method) are natural candidate features for such a model.